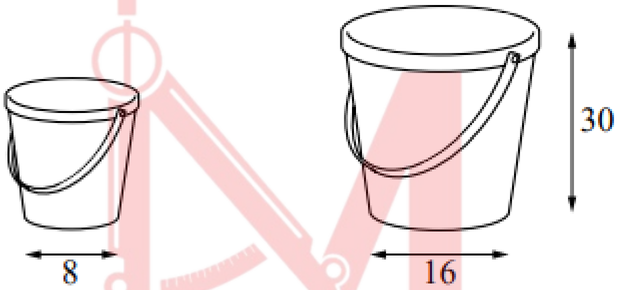
Similar Triangles and Figures

Worksheet 2 (Non-Calculator)

1. S09/qp1/q11

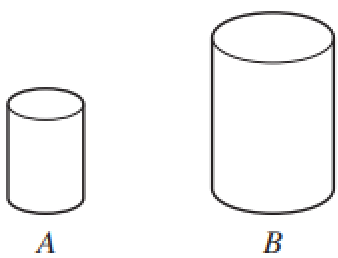
Similar buckets are available in two sizes. The large bucket has height 30 cm and base diameter 16 cm. The small bucket has base diameter 8 cm.



a. Find the height of the small bucket. [1]

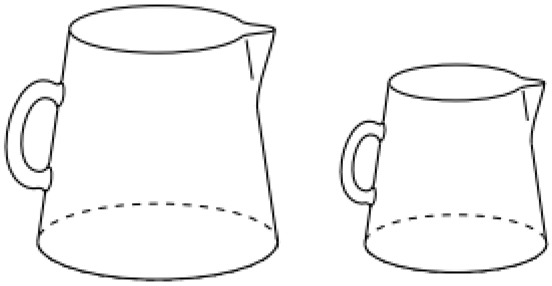
b. Given that the small bucket has volume $850\ cm^3$, find the volume of the large bucket. [2]

2. S11/qp12/q10



These two cylinders are similar. A B The ratio of their volumes is 8 : 27. The height of cylinder A is 12 cm. Find the height of cylinder B. [2]

3. W12/qp21/q7(c)



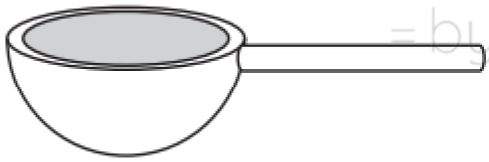
Two geometrically similar jugs have volumes of $1000\ cm^3$ and $512\ cm^3$. They have circular bases. The diameter of the base of the larger jug is 9 cm. Calculate the diameter of the base of the smaller jug. [3]

4. W14/qp12/q8

Two bottles are geometrically similar. The ratio of the areas of their bases is 1 : 4. Write down the ratios of their

a. heights, [1]
b. volumes. [1]

5. W15/qp22/q4(ii)
You may use a calculator here



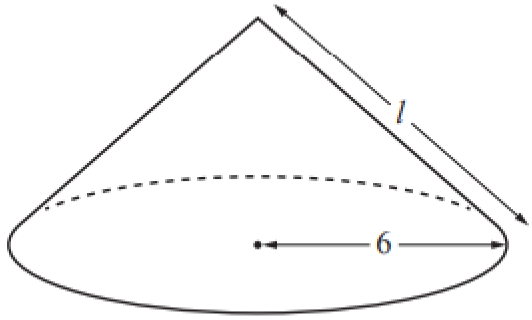
A spoon used for measuring in cookery consists of a hemispherical bowl and a handle. The internal volume of the hemispherical bowl is $20\ cm^3$. The handle is of length 5cm.

The hemispherical bowl of a geometrically similar spoon has an internal volume of $50cm^3$. Find the length of its handle. [2]

6. S21/qp22/q9(c)

[Volume of a cone $= \frac{1}{3}\pi r^2 h$]

[Curved surface area of a cone $= \pi r l$]



The total surface area of the cone above is $84\pi\ cm^2$.

A similar cone has a total surface area of $47.25\pi\ cm^2$.

Find the radius of this cone. [2]

Answer Key

1. S09/qp1/q11

11	(a)	15	1	
	(b)	6 800	2 *	Ratio of corresponding lengths **cubed** soi

2. S11/qp12/q10

10	18	2	B1 for attempt at $\sqrt[3]{8}$: $\sqrt[3]{27}$ or M1 for $12^3 : x^3 = 8 : 27$ oe

3. W12/qp21/q7(c)

(c) 7.2	3	**M2** for $\frac{x}{9} = \sqrt[3]{\frac{512}{1000}}$ oe or **B1** for $\sqrt[3]{512}$: $\sqrt[3]{1000}$ soi

4. W14/qp12/q8

8	(a)	1 : 2 oe	**1**	
	(b)	1 : 8 oe, or ft *their*(a) cubed	**1**√	

5. W15/qp22/q4(ii)

(ii)	6.79	2	**B1** for $\sqrt[3]{\frac{50}{20}}$ or $\sqrt[3]{\frac{20}{50}}$ oe or **M1** for $\left(\frac{5}{x}\right)^3 = \frac{20}{50}$ oe

6. S21/qp22/q9(c)

9(c)	4.5 nfww	2	**B1** for $\sqrt{\frac{47.25\pi}{84\pi}}$ soi or $\sqrt{\frac{84\pi}{47.25\pi}}$ soi or **M1** for a correct equation in *r*